

SAT Math Mastery · Week 9A

Solids, Coordinate Geometry & Composite Shapes

Time	150 minutes: 110 min solving + 40 min self-check and error log
Topics	volume and surface area of solids · scaling laws · composite, glued and inscribed solids · distance, midpoint, slope · area from coordinates · composite and shaded regions · rate of fill and unit volume · dilations
Name / Date	_____

1 · SKILLS & IDEAS

- S1.** Volume of prisms, cylinders, cones, spheres and pyramids; recover a radius from a circumference, a volume or a surface area.
- S2.** Surface area: total vs lateral, slant height of a cone or pyramid, surface area of a prism from its three face pairs.
- S3.** Scaling laws: lengths $\times k$, areas $\times k^2$, volumes $\times k^3$; recover k from an area or volume ratio.
- S4.** Composite, glued, removed and inscribed solids (sphere in a cube, cylinder in a sphere, joined prisms hide two faces).
- S5.** Coordinate geometry: distance, midpoint (including a missing endpoint), slope, equidistant points.
- S6.** Area from coordinates: base and height on a grid, decomposition, and the shoelace formula.
- S7.** Composite and shaded plane regions: subtract a part from a whole; equilateral-triangle trapezoids; algebraic side lengths leading to a quadratic.
- S8.** Rate of fill, unit-volume conversions ($\text{m}^3 \leftrightarrow \text{L}$), and dilations in the coordinate plane.

2 · REQUIRED FACTS

- Volume of any prism or cylinder is $V = Bh$ (area of the base times the height); a cone or pyramid with the same base and height holds exactly $\frac{1}{3}$ of that.
- The slant height ℓ , the height h and the base radius r of a right cone form a right triangle: $\ell^2 = r^2 + h^2$.
- A cube's space diagonal is $s\sqrt{3}$; a rectangular box's is $\sqrt{a^2 + b^2 + c^2}$. A rectangle inscribed in a circle has its diagonal equal to the diameter.
- When two solids are glued along a face, the surface area is the sum of the two surface areas minus *twice* the area of the glued face.
- Similar figures: if lengths scale by k , areas scale by k^2 and volumes by k^3 ; angle measures never change.
- $1 \text{ m}^3 = 1000 \text{ L}$ and $1 \text{ cm}^3 = 1 \text{ mL}$; square and cubic units convert with the *square* and *cube* of the length factor.
- A dilation centered at the origin with scale factor k sends $(x, y) \rightarrow (kx, ky)$, multiplies lengths by k and areas by k^2 .

3 · REQUIRED FORMULAS

$$\begin{aligned}
 V_{\text{prism}} &= Bh, & V_{\text{cyl}} &= \pi r^2 h, & V_{\text{cone}} &= \frac{1}{3} \pi r^2 h, & V_{\text{sphere}} &= \frac{4}{3} \pi r^3, & V_{\text{pyramid}} &= \frac{1}{3} Bh \\
 SA_{\text{sphere}} &= 4\pi r^2, & SA_{\text{cyl}} &= 2\pi r^2 + 2\pi rh, & SA_{\text{cone}} &= \pi r^2 + \pi r\ell, & SA_{\text{box}} &= 2(ab + bc + ac) \\
 d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}, & M &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right), & \text{Area} &= \frac{1}{2} \left| \sum (x_i y_{i+1} - x_{i+1} y_i) \right|
 \end{aligned}$$

4 · MINI THEORY SHEET

Choose the solid's key quantity first.

Given	Fastest route
a circumference C	$r = \frac{C}{2\pi}$, then use r^2 in the area or volume
radius and slant height of a cone	$h = \sqrt{\ell^2 - r^2}$ (look for 6-8-10, 5-12-13)
a volume $= k\pi$	divide by π first; solve r^3 or r^2h with whole-number guesses
lateral area of a pyramid	each triangle $= \frac{1}{2}(\text{side})\ell$; find ℓ , then $h = \sqrt{\ell^2 - (\text{side}/2)^2}$
a rate of fill Q per minute	height rate $= \frac{Q}{\text{base area}}$ (not divided by the radius)

Inscribed solids. Draw the cross section: a sphere in a cube touches the faces (diameter = edge); a sphere around a cube passes through the vertices (diameter = space diagonal); a cylinder in a sphere has the sphere's radius as the hypotenuse of a right triangle with the half-height and the base radius.

Scaling. If the ratio of areas is $a : b$, then $k = \sqrt{b/a}$; if the ratio of volumes is $a : b$, then $k = \sqrt[3]{b/a}$. Convert, then apply to the quantity asked.

Area from coordinates. Axis-parallel side $\rightarrow$ base and height. Otherwise use the shoelace sum, or box the figure and subtract corner triangles.

Unit volume. Convert every quantity to one unit *before* dividing; a rate in liters per minute needs a volume in liters.

Speed shortcuts.

Situation	Shortcut
ratio of volumes $a : b$	lengths $\sqrt[3]{b/a}$, areas $\left(\sqrt[3]{b/a}\right)^2$
radius $\times m$, height $\times n$	volume $\times m^2n$ (a cylinder or cone)
cube with surface area SA	face $= SA/6$, edge $= \sqrt{SA/6}$
glued prisms	$SA = SA_1 + SA_2 - 2(\text{glued face})$, or rebuild the combined box
missing endpoint	$B = 2M - A$ coordinatewise
cone or sphere with a π in the volume	cancel π at once and solve for r
solid of revolution of a right triangle	cone volume $\frac{1}{3}\pi r^2 h$ with r the distance to the axis
dilation by k about the origin	multiply every coordinate by k ; area $\times k^2$

5 · GUIDED EXAMPLES

Example 1 — Composite solid: cylinder topped by a hemisphere

A silo is a cylinder of radius 3 and height 10 with a hemisphere of radius 3 on top. Find its volume.

Cylinder: $\pi r^2 h = \pi(9)(10) = 90\pi$. Hemisphere: $\frac{1}{2} \cdot \frac{4}{3}\pi r^3 = \frac{2}{3}\pi(27) = 18\pi$.

Total volume $= 90\pi + 18\pi = 108\pi$.

Example 2 — Scaling: area ratio to volume

Two similar prisms have surface areas 50 and 162. The smaller has volume 125. Find the volume of the larger.

Area ratio $\frac{162}{50} = \frac{81}{25} = k^2$, so $k = \frac{9}{5}$. Volume ratio $k^3 = \frac{729}{125}$.

Larger volume $= 125 \cdot \frac{729}{125} = 729$.

Example 3 — Area from coordinates (shoelace)

Find the area of the triangle with vertices $P(-1, 2)$, $Q(5, -2)$, $R(3, 6)$.

$\sum (x_i y_{i+1} - x_{i+1} y_i)$: $P \rightarrow Q$: $(-1)(-2) - (5)(2) = -8$; $Q \rightarrow R$: $(5)(6) - (3)(-2) = 36$; $R \rightarrow P$: $(3)(2) - (-1)(6) = 12$.

Sum $= -8 + 36 + 12 = 40$, so area $= \frac{1}{2}|40| = 20$.

6 · PRACTICE SET — 44 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

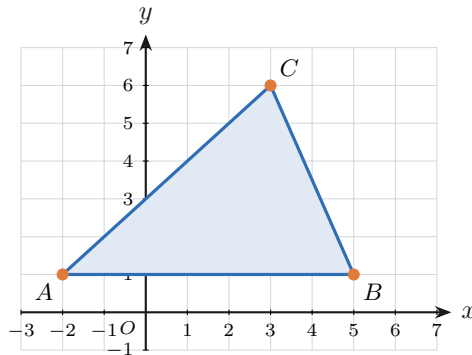
1. EASY · NO CALCULATOR NEEDED

A right circular cylinder has a height of 5 and a base with circumference 8π . What is the volume of the cylinder?

- (A) 40π (B) 80π
(C) 160π (D) 320π
- _____
- _____
- _____

2. EASY · NO CALCULATOR NEEDED

In the xy -plane, the vertices of triangle ABC are plotted as shown. What is the area of triangle ABC ?



3. EASY · NO CALCULATOR NEEDED

Point $M(3, 5)$ is the midpoint of $\overline{AB}$. If A has coordinates $(-1, 2)$, what are the coordinates of B ?

- (A) $\left(1, \frac{7}{2}\right)$ (B) $(8, 7)$
(C) $(7, 8)$ (D) $(4, 3)$
- _____
- _____
- _____

4.

EASY · CALCULATOR

A cone has a volume of 14 cubic centimeters. A second cone is similar to the first, and each linear dimension of the second cone is 3 times the corresponding dimension of the first. What is the volume, in cubic centimeters, of the second cone?

5.

EASY · NO CALCULATOR NEEDED

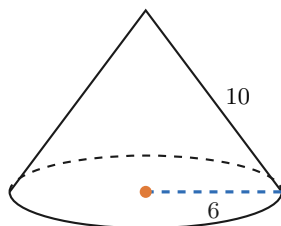
A cube has a total surface area of 150 square inches. What is the volume of the cube, in cubic inches?

- (A) 125 (B) 150
(C) 625 (D) 25

6.

MEDIUM · CALCULATOR

The cone shown has a base radius of 6 and a slant height of 10. What is the volume of the cone?



- (A) 120π (B) 288π
(C) 360π (D) 96π

7.

MEDIUM · NO CALCULATOR NEEDED

A sphere has a volume of 288π cubic centimeters. The surface area of the sphere is $k\pi$ square centimeters. What is the value of k ?

8.

MEDIUM · NO CALCULATOR NEEDED

A cylinder has radius r and height h . Which of the following changes to the cylinder results in a volume that is exactly twice the original volume?

I. Multiply the radius by $\sqrt{2}$ and leave the height unchanged.

II. Multiply the radius by 2 and divide the height by 2.

III. Multiply the radius by 2 and divide the height by 4.

(A) I only

(B) II only

(C) I and II only

(D) I, II, and III

9.

MEDIUM · NO CALCULATOR NEEDED

A rectangular box has a square base with side length 6 and a height of h . The total surface area of the box is 240 square units. What is the value of h ?

10.

MEDIUM · NO CALCULATOR NEEDED

The endpoints of a diameter of a circle in the xy -plane are $(-3, 2)$ and $(5, 8)$. What is the radius of the circle?

- (A) 100 (B) 5
(C) 10 (D) 25
-
-
-
-
-
-
-

11.

MEDIUM · NO CALCULATOR NEEDED

Triangle ABC is similar to triangle DEF . The area of triangle ABC is 48 and the area of triangle DEF is 108. The longest side of triangle ABC has length 10. What is the length of the longest side of triangle DEF ?

12.

MEDIUM · CALCULATOR

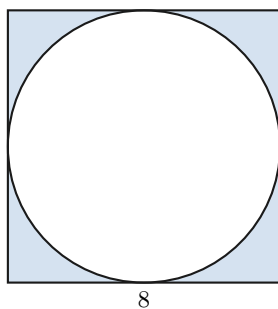
A cylindrical tank has an inside radius of 2 meters. Water flows into the tank at a constant rate of 0.5 cubic meter per minute. Which of the following is closest to the rate, in meters per minute, at which the water level in the tank rises?

- (A) 0.08 (B) 0.16
(C) 0.25 (D) 0.04
-
-
-
-
-
-
-

13.

MEDIUM · NO CALCULATOR NEEDED

A circle is inscribed in a square with side length 8, as shown. What is the area of the shaded region?



(A) $64 - 16\pi$

(B) $64 - 8\pi$

(C) $64 - 4\pi$

(D) 16π

14.

HARD · CALCULATOR

A solid sphere with a radius of 3 centimeters is placed in a cylindrical container that holds water and has an inside base radius of 5 centimeters. The sphere is completely submerged and no water spills. By how many centimeters does the water level rise?

15.

HARD · NO CALCULATOR NEEDED

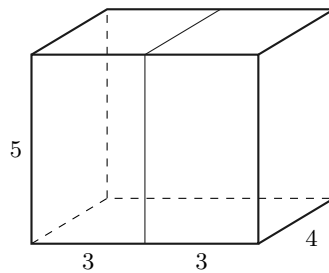
All eight vertices of a cube with edge length 6 lie on the surface of a sphere. What is the volume of the sphere?

- (A) $72\sqrt{2}\pi$ (B) $108\sqrt{3}\pi$
(C) 288π (D) 36π
-
-
-
-
-
-
-
-

16.

HARD · NO CALCULATOR NEEDED

Two identical rectangular blocks, each measuring 3 by 4 by 5, are glued together along a 4-by-5 face to form one larger rectangular block, as shown. What is the total surface area of the larger block?



- (A) 188 (B) 94
(C) 148 (D) 168
-
-
-
-
-
-
-
-

17.

HARD · CALCULATOR

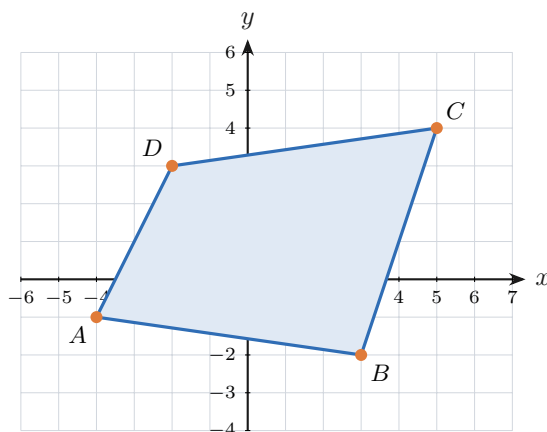
The four lateral faces of a right square pyramid are congruent triangles with a combined area of 260. The base of the pyramid is a square with side length 10. What is the volume of the pyramid?

- (A) 400 (B) 1200
(C) 1300 (D) $\frac{1300}{3}$

18.

HARD · CALCULATOR

In the xy -plane, quadrilateral $ABCD$ has the vertices shown on the grid. What is the area of quadrilateral $ABCD$?



- (A) 70 (B) $\frac{63}{2}$
(C) 38 (D) 35

19.

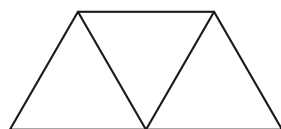
HARD · NO CALCULATOR NEEDED

A rectangle with side lengths 10 and 24 is inscribed in a circle. The area of the circle is $k\pi$. What is the value of k ?

20.

HARD · CALCULATOR

The trapezoid shown is composed of three congruent equilateral triangles. The perimeter of the trapezoid is 20. What is the area of the trapezoid?

(A) $48\sqrt{3}$ (B) $4\sqrt{3}$ (C) $12\sqrt{3}$ (D) $16\sqrt{3}$

21.

HARD · CALCULATOR

A rectangular pool measures 6 meters by 4 meters and is 1.5 meters deep. A pump fills the empty pool at a constant rate of 150 liters per minute. How many hours does it take to fill the pool completely? (1 cubic meter = 1000 liters)

22.

HARD · NO CALCULATOR NEEDED

Two solid cones are similar. The volume of the larger cone is $\frac{27}{8}$ times the volume of the smaller cone. The surface area of the smaller cone is 100 square centimeters. What is the surface area, in square centimeters, of the larger cone?

(A) 150

(B) 225

(C) 337.5

(D) 800

23.

HARD · CALCULATOR

In the xy -plane, triangle ABC has vertices $A(1, 2)$, $B(7, 2)$ and $C(1, 10)$. Triangle DEF is the image of triangle ABC under a dilation centered at the origin, where D is the image of A and $D = (3, 6)$. What is the area of triangle DEF ?

(A) 648

(B) 72

(C) 144

(D) 216

24.

SAT CHALLENGE · CALCULATOR

A cone-shaped paper cup is held point down. The cup has an inside radius of 6 centimeters at the top and a height of 12 centimeters. The cup contains water to a depth of 8 centimeters. All of this water is poured into an empty cylindrical glass with an inside radius of 4 centimeters. What is the depth, in centimeters, of the water in the glass?

25.

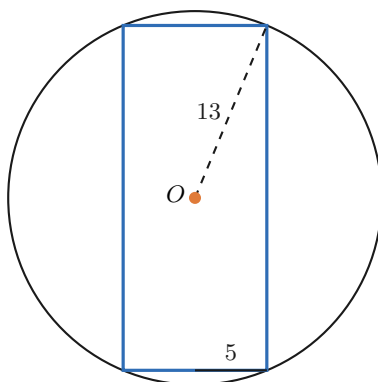
SAT CHALLENGE · DESMOS

In the xy -plane, point P lies on the line $y = 2x + 1$ and is the same distance from $A(-1, 6)$ as from $B(5, 0)$. What is the area of triangle ABP ?

26.

SAT CHALLENGE · CALCULATOR

A right circular cylinder is inscribed in a sphere with radius 13 so that the circumference of each base of the cylinder lies on the sphere. The base of the cylinder has radius 5. The figure shows a cross section through the center O of the sphere. What is the volume of the cylinder?

(A) 600π (B) 650π (C) 300π (D) 325π

27.

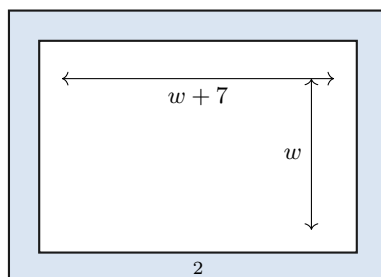
SAT CHALLENGE · NO CALCULATOR NEEDED

A scale model of a water tank holds 8 liters of water. The actual tank holds 27 cubic meters of water, and the model is geometrically similar to the actual tank. The model is 36 centimeters tall. How tall, in meters, is the actual tank? (1 cubic meter = 1000 liters)

28.

SAT CHALLENGE · CALCULATOR

A rectangular garden has a width of w meters and a length that is 7 meters greater than its width. A path of uniform width 2 meters surrounds the garden on all sides. The area of the path alone is 156 square meters. What is the area, in square meters, of the garden?



(A) 98

(B) 294

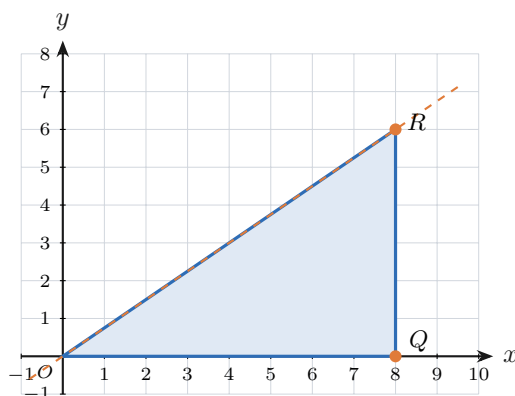
(C) 450

(D) 588

29.

SAT CHALLENGE · CALCULATOR

In the xy -plane, triangle OQR has vertices $O(0,0)$, $Q(8,0)$ and $R(8,6)$, as shown. The triangle is rotated 360° about the line containing side $\overline{OR}$ to form a solid. What is the volume of the solid?



(A) 128π

(B) $\frac{1152}{5}\pi$

(C) $\frac{384}{5}\pi$

(D) 96π

30.

SAT CHALLENGE · CALCULATOR

A solid metal sphere with radius 9 centimeters is melted. Exactly 90% of the metal is recast into identical solid cones, each with radius 3 centimeters and height 4 centimeters. What is the greatest number of complete cones that can be made?

31.

SAT CHALLENGE • NO CALCULATOR NEEDED

Two similar rectangular crates have surface areas of 144 square feet and 400 square feet. The smaller crate has a volume of 108 cubic feet. What is the volume, in cubic feet, of the larger crate?

- (A) 500 (B) 1500
(C) 180 (D) 300

32.

SAT CHALLENGE · NO CALCULATOR NEEDED

Two similar solid blocks have surface areas of 208 square centimeters and 325 square centimeters. The larger block has a volume of 375 cubic centimeters. What is the volume, in cubic centimeters, of the smaller block?

33.

SAT CHALLENGE · CALCULATOR

A closed right circular cylinder has a volume of 48π cubic centimeters and a total surface area of 56π square centimeters. The height of the cylinder is greater than its diameter. What is the height, in centimeters, of the cylinder?

(A) 2

(B) 3

(C) 4

(D) 12

34.

SAT CHALLENGE · NO CALCULATOR NEEDED

Sphere A has a great circle with circumference 8π centimeters. The surface area of sphere B is 80π square centimeters greater than the surface area of sphere A . What is the ratio of the volume of sphere B to the volume of sphere A ?

(A) $\frac{3}{2}$
(C) $\frac{27}{8}$

(B) $\frac{9}{4}$
(D) $\frac{5}{4}$

35.

SAT CHALLENGE · CALCULATOR

A right pyramid has a square base. The diagonal of the base is $10\sqrt{2}$ centimeters and the volume of the pyramid is 400 cubic centimeters. What is the lateral surface area, in square centimeters, of the pyramid (the four triangular faces, not the base)?

(A) 360

(B) 260

(C) 240

(D) 130

36.

SAT CHALLENGE • CALCULATOR

A metal pipe is a hollow cylinder of length 15 centimeters and outer radius 10 centimeters. The wall has a uniform thickness of t centimeters, where $0 < t < 10$. The volume of metal in the pipe is 765π cubic centimeters. What is the value of t ?

37.

SAT CHALLENGE • NO CALCULATOR NEEDED

A right pyramid has a rectangular base. Its height is 12 centimeters, and the slant heights of its triangular faces are 13 centimeters for one pair of opposite faces and 20 centimeters for the other pair. What is the total surface area, in square centimeters, of the pyramid, including the base?

- (A) 1160 (B) 616
(C) 866 (D) 936

38.

SAT CHALLENGE · CALCULATOR

A raised garden bed is filled with soil. A vertical cross-section of the soil, perpendicular to the 12-foot length of the bed, is a trapezoid: the bed is 8 feet wide, and the soil is 1.5 feet deep along one side and 3.5 feet deep along the opposite side, with the surface sloping uniformly between them. Soil is sold at \$45 per cubic yard and billed for the exact volume (1 yard = 3 feet). What is the cost, in dollars, of the soil needed?

39.

SAT CHALLENGE · CALCULATOR

Cylinder P has radius 2 centimeters and height 6 centimeters. Cylinder Q is similar to cylinder P and has a total surface area of 200π square centimeters. What is the volume, in cubic centimeters, of cylinder Q ?

(A) 375π (B) 600π (C) 60π (D) 150π

40.

SAT CHALLENGE · CALCULATOR

Rectangular box P has dimensions 3 inches by 4 inches by 6 inches. Box Q is similar to box P and has a volume of 1125 cubic inches. What is the total surface area, in square inches, of box Q ?

41.

SAT CHALLENGE · NO CALCULATOR NEEDED

A sphere is inscribed in a right circular cone whose base has radius 5 and whose height is 12, so that the sphere is tangent to the base and to the lateral surface of the cone. What fraction of the volume of the cone is occupied by the sphere? (Enter your answer as a fraction.)

42.

SAT CHALLENGE · CALCULATOR

A museum has a small solid model of a sculpture and a full-size copy; the two are similar solids. The sealant needed to coat the whole surface is proportional to surface area. The model needs 1.6 liters of sealant and the full-size copy needs 14.4 liters. The model has a volume of 2.5 cubic feet. What is the volume, in cubic feet, of the full-size copy?

(A) 22.5

(B) 67.5

(C) 202.5

(D) 7.5

43.

SAT CHALLENGE · NO CALCULATOR NEEDED

Two aquariums are similar in shape. The smaller aquarium holds 250 liters and has 400 square decimeters of glass in total. The larger aquarium holds 432 liters. How many square decimeters of glass does the larger aquarium have in total? (Note that 1 liter = 1 cubic decimeter.)

44.

SAT CHALLENGE · NO CALCULATOR NEEDED

A roof ornament is a solid pyramid whose base is a square of area 100 square inches. The apex is directly above the center of the base, and the volume of the pyramid is 400 cubic inches. What is the combined area, in square inches, of the four triangular lateral faces?

(A) 130

(B) 360

(C) 260

(D) 240